RNA-seq → ceRNA Pipeline — Tools & Documentation Study Guide

diabetes-cerna-rnaseq · a learner's reading list

How to use this guide

This lists every tool in the pipeline, grouped by stage, with:

- **Does** — what the tool is for (the concept).
- **Its job here** — what it does in *this* project.
- **Learn these options** — the few flags/settings actually worth understanding.
- **Docs** — where to read more.

Priority key (how much to invest in reading its documentation):

- [**Must read**] — core tool; understanding its options builds real intuition.
- [**Good to read**] — worth skimming to understand what it reports/does.
- [**Reference**] — use as needed; don't study deeply.

How to read any tool's docs (reminder): run `toolname --help` in the terminal for the flag list, or search "toolname documentation" online. Don't read cover-to-cover — skim the summary, study the **options list** (each flag says what it does and when to use it), and copy the **Examples** section.

Stage 1 — Getting the data

sra-tools (prefetch, fasterq-dump) — [Reference]

- **Does:** downloads raw sequencing files (FASTQ) from NCBI's SRA archive.
- **Its job here:** pulled your 12 FASTQ samples from SRA.
- **Learn these options:** `--split-files` (paired vs single output), `--threads`; `prefetch` then `fasterq-dump` as a two-step pattern.
- **Docs:** https://github.com/ncbi/sra-tools/wiki

pysradb — [Reference]

- **Does:** converts between the different accession ID systems (GSM, SRR, SRX…).
- **Its job here:** mapped your GSM sample IDs to SRR run IDs for download.
- **Learn these options:** the `gsm-to-srr` subcommand; `--saveto`.
- **Docs:** https://saketkc.github.io/pysradb/

Stage 2 — Quality control & trimming

FastQC — [Good to read]

- **Does:** inspects raw read quality and produces per-file reports.
- **Its job here:** the "health check" on your reads, before and after trimming.
- **Learn these options:** mostly about *interpreting* its modules (quality, adapter, duplication, GC) — see your `how_to_read_qc.md`. Few tunable flags.
- **Docs:** https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

fastp — [Must read]

- **Does:** all-in-one read cleaner — trims adapters, low-quality bases, short reads.
- **Its job here:** cleaned all 12 samples; this is where trimming decisions live.
- **Learn these options:** `--qualified_quality_phred` (quality cutoff), `--length_required` (min length), adapter auto-detection, single vs paired input (`-i/-o` vs `-i/-I/-o/-O`), `--detect_adapter_for_pe`.
- **Docs:** https://github.com/OpenGene/fastp

MultiQC — [Good to read]

- **Does:** merges many tool reports (FastQC, fastp, HISAT2, featureCounts…) into one combined dashboard.
- **Its job here:** gave you the single before/after QC and alignment reports.
- **Learn these options:** `-o` (output dir), `-n` (report name); understanding which tools it auto-detects.
- **Docs:** https://multiqc.info/docs/

Stage 3 — Alignment & counting

HISAT2 — [Must read]

- **Does:** aligns reads to a reference genome (finds where each read came from).
- **Its job here:** mapped your cleaned reads to GRCh38 → BAM files.
- **Learn these options:** `-x` (index), `-U` vs `-1/-2` (single vs paired), `-p` (threads), `--new-summary`; the idea of a prebuilt **index**.
- **Docs:** https://daehwankimlab.github.io/hisat2/manual/

samtools — [Must read]

- **Does:** the Swiss-army knife for SAM/BAM alignment files (sort, index, view, stats).
- **Its job here:** sorted and indexed your BAMs after alignment.
- **Learn these options:** `sort`, `index`, `view` (esp. `-H` for the header), `flagstat`; what a BAM/BAI is.
- **Docs:** http://www.htslib.org/doc/samtools.html

featureCounts (Subread) — [Must read]

- **Does:** counts how many reads land on each gene → the count matrix.
- **Its job here:** turns your BAMs into a gene × sample count table (Day 5).
- **Learn these options:** `-a` (GTF annotation), `-s` (strandedness: 0/1/2), `-p --countReadPairs` (paired only), `-T` (threads); why multimapped reads are dropped.
- **Docs:** https://subread.sourceforge.net/SubreadUsersGuide.pdf

Stage 4 — Differential expression & annotation (R)

DESeq2 — [Must read]

- **Does:** finds genes reliably different between groups (statistics).
- **Its job here:** ND vs T2D differential expression on the published matrix.
- **Learn these options:** the `design = ~ condition` formula, `results(contrast=…)`, low-count filtering, `padj` vs `pvalue`, `log2FoldChange`, `vst()` for PCA.
- **Docs:** https://bioconductor.org/packages/release/bioc/vignettes/DESeq2/inst/doc/DESeq2.html

biomaRt — [Good to read]

- **Does:** looks up gene annotation (symbols, biotypes) from Ensembl.
- **Its job here:** labels each gene as protein_coding / lncRNA (the biotype split).
- **Learn these options:** `useEnsembl()`, `getBM()` (attributes/filters/values).
- **Docs:** https://bioconductor.org/packages/release/bioc/vignettes/biomaRt/inst/doc/accessing_ensembl.htı

clusterProfiler — [Good to read]

- **Does:** enrichment analysis — finds common themes (GO terms, KEGG pathways) in a gene list.
- **Its job here:** the biological story from your DE mRNAs.
- **Learn these options:** `enrichGO()`, `enrichKEGG()`, `ont`, `pAdjustMethod`, `qvalueCutoff`.
- **Docs:** https://yulab-smu.top/biomedical-knowledge-mining-book/

org.Hs.eg.db — [Reference]

- **Does:** a human gene annotation database (ID conversions).
- **Its job here:** maps gene symbols to Entrez IDs for enrichment.
- **Docs:** https://bioconductor.org/packages/org.Hs.eg.db/

EnhancedVolcano — [Reference]

- **Does:** draws volcano plots of DE results.
- **Its job here:** the volcano figure (up/down genes).
- **Learn these options:** `pCutoff`, `FCcutoff`, `x`, `y`.
- **Docs:** https://bioconductor.org/packages/EnhancedVolcano/

pheatmap — [Reference]

- **Does:** draws clustered heatmaps.
- **Its job here:** heatmap of top DE genes.
- **Docs:** https://cran.r-project.org/package=pheatmap

Stage 5 — miRNA prediction & ceRNA network

multiMiR — [Good to read]

- **Does:** queries miRNA–target databases (miRTarBase, TargetScan, miRDB…).
- **Its job here:** predicts miRNA→mRNA edges for the ceRNA network.
- **Learn these options:** `get_multimir()`, `table = "validated"` vs `"predicted"`, `org`.
- **Docs:** https://bioconductor.org/packages/multiMiR/

igraph — [Reference]

- **Does:** builds and analyzes networks (nodes + edges).
- **Its job here:** assembles and draws the lncRNA–miRNA–mRNA ceRNA network.
- **Learn these options:** `graph_from_data_frame()`, `degree()`, layout functions.
- **Docs:** https://r.igraph.org/

Databases (not tools — data sources)

- **ENCORI / starBase** — lncRNA<->miRNA interactions: https://rnasysu.com/encori/
- **miRcode** — predicted miRNA targets: http://www.mircode.org/

Supporting tools (whole project)

conda / Miniforge — [Good to read]

- **Does:** manages software environments so your setup is reproducible.
- **Learn these options:** `create`, `activate`, `install`, channels (`-c bioconda`), `env export`; the `CONDA_SUBDIR=osx-64` trick for Apple Silicon.
- **Docs:** https://docs.conda.io/

git / GitHub — [Good to read]

- **Does:** version control — tracks changes and hosts your code publicly.
- **Learn these options:** `add`, `commit`, `push`, `status`, `.gitignore`.
- **Docs:** https://git-scm.com/doc

Suggested reading order (start here)

If you only deeply read a handful, do these five — they carry the most transferable intuition:

1. **fastp** — what "cleaning reads" really means and what you can tune.
2. **HISAT2** — how alignment and genome indexes work.
3. **featureCounts** — how reads become gene counts (strandedness matters!).
4. **DESeq2** — the statistics of "differentially expressed."
5. **samtools** — the universal language of alignment files.

Then skim the [**Good to read**] ones as you reach their step. Everything marked [**Reference**] you can just look up when needed.

The habit that builds independence: before running any command I give you, run `toolname --help`, find each flag in the output, and match it to what it does. That's how the commands stop being magic and start being *yours*.